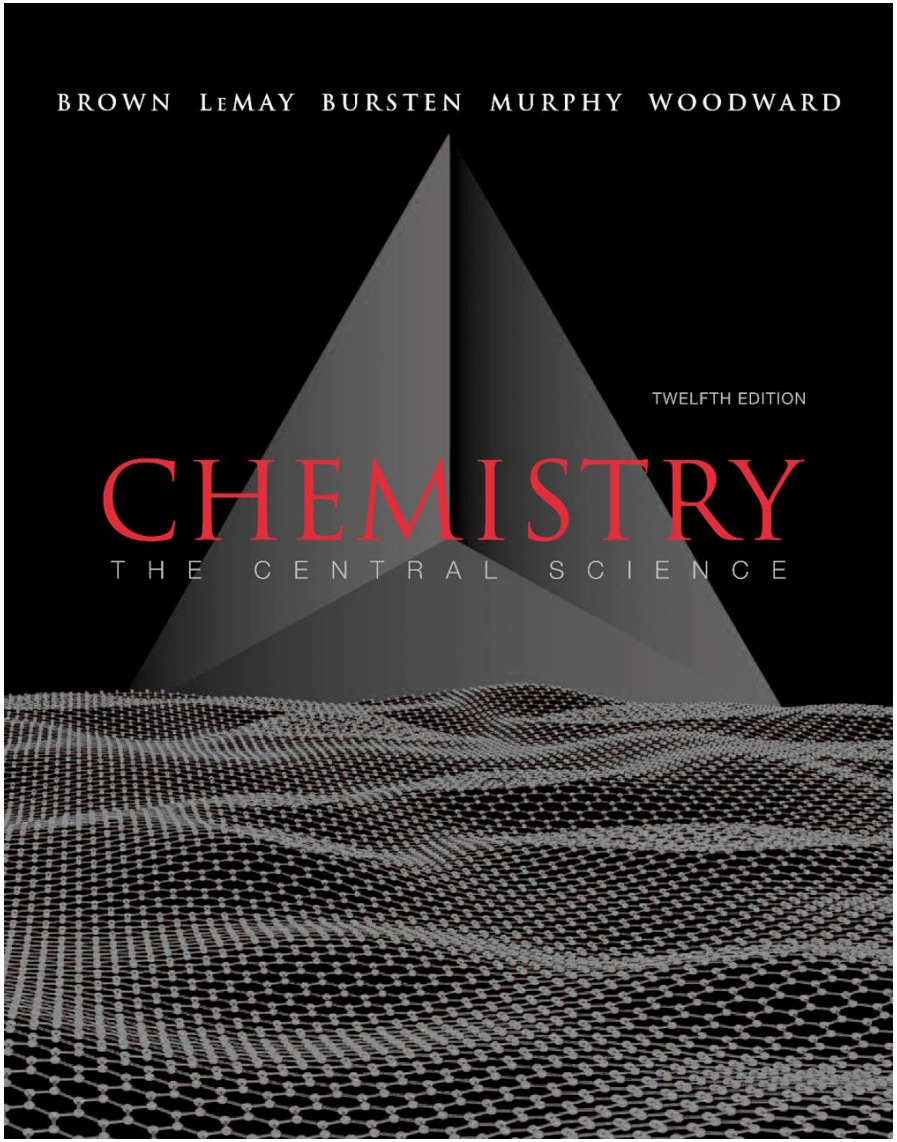


Lecture Presentation

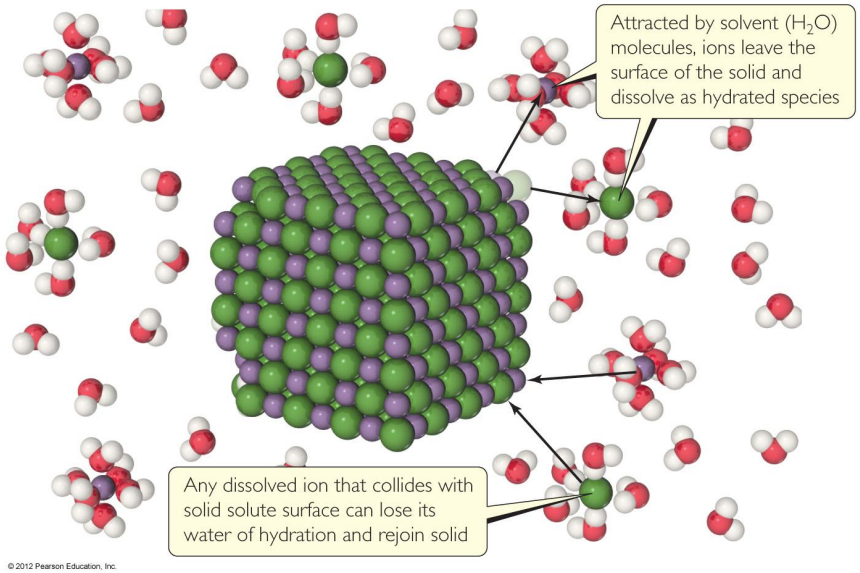
Chapter 13

Properties of Solutions

John D. Bookstaver
St. Charles Community College
Cottleville, MO

© 2012 Pearson Education, Inc.

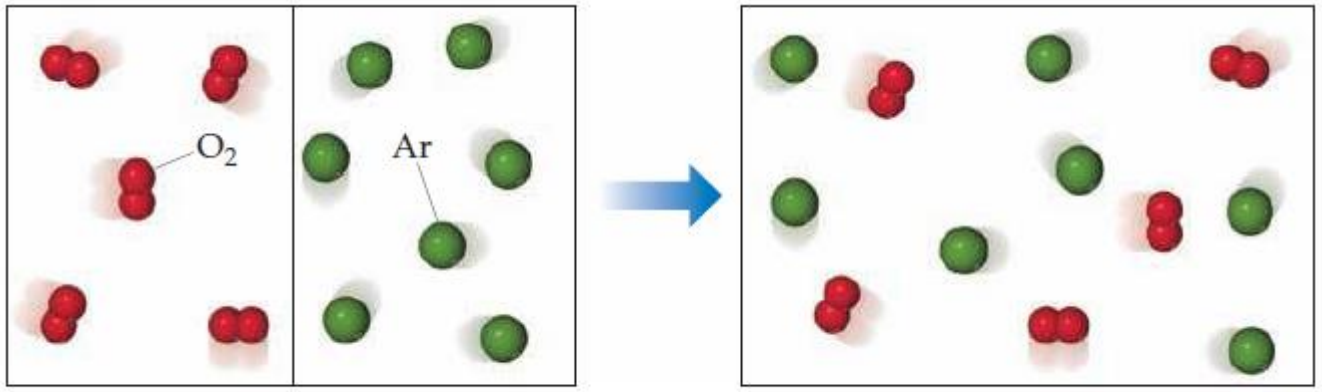
Solutions



- **Solutions** are homogeneous mixtures of two or more pure substances.
- In a solution, the **solute** is dispersed uniformly throughout the **solvent**.

© 2012 Pearson Education, Inc.

The Natural Tendency toward Mixing



▲ Figure 13.1 Spontaneous mixing of two gases to form a homogeneous mixture (a solution).

The mixing of gases is a spontaneous process, meaning it occurs of its own accord without any input of energy from outside the system. When the molecules mix and become more randomly distributed, there is an increase in a thermodynamic quantity called entropy.

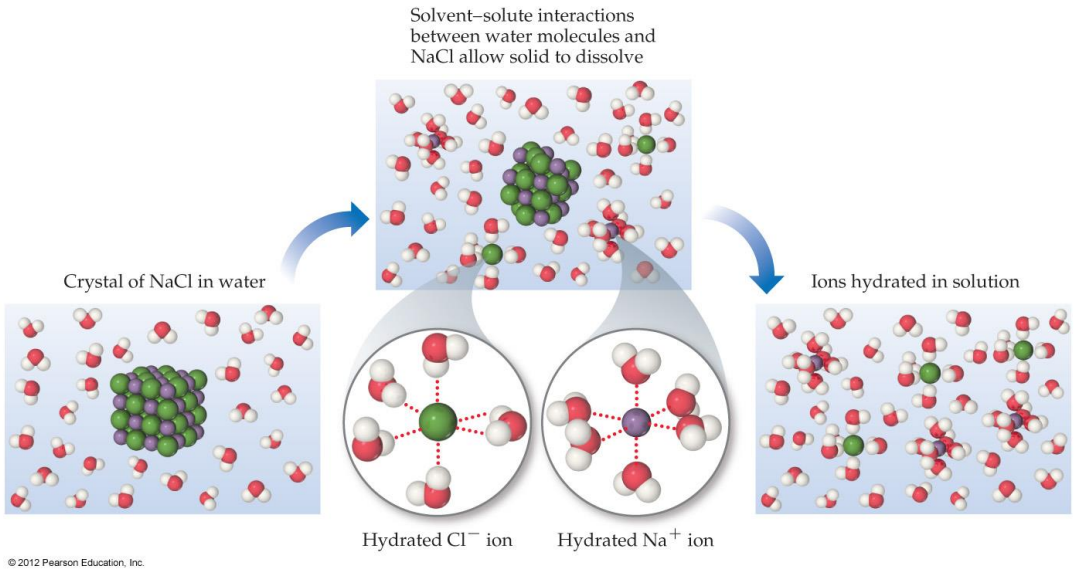
When molecules of different types are brought together, mixing occurs spontaneously
unless the molecules are restrained either by sufficiently strong intermolecular
forces or by physical barriers.
Thus, gases spontaneously mix unless restrained
by their containers because with gases intermolecular forces are too weak to restrain
the molecules.

However, when the solvent or solute is a solid or liquid, intermolecularforces become important in determining whether a solution forms.

For example,
although ionic bonds hold sodium and chloride ions together in solid sodium chloride, the solid dissolves in water because of the compensating strength of the attractive forces between the ions and water molecules.

Sodium chloride does not dissolve in gasoline, however, because the intermolecular forces between the ions and the gasoline molecules are too weak.

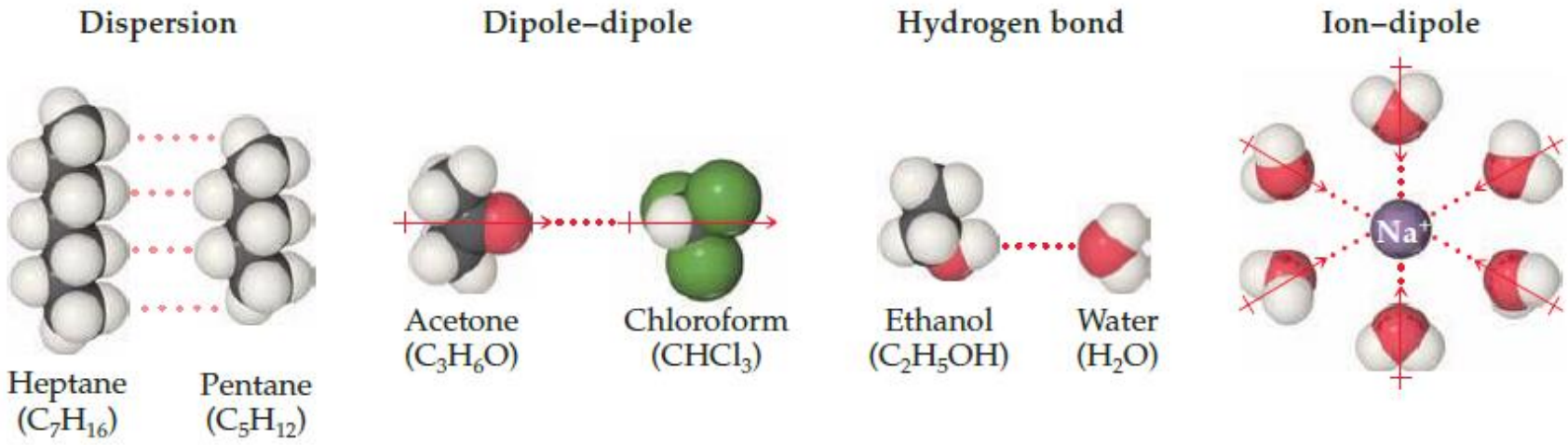
The Effect of Intermolecular Forces on Solution Formation



The intermolecular forces between solute and solvent particles must be strong enough to compete with those between solute particles and those between solvent particles.

© 2012 Pearson Education, Inc.

Any of the intermolecular forces discussed in Chapter 11 can operate between solute and solvent particles in a solution. These forces are summarized in ▶ Figure 13.2. Dispersion forces, for example, dominate when one nonpolar substance, such as C_7H_{16}, dissolves in another, such as C_5H_{12}, and ion–dipole forces dominate in solutions of ionic substances in water.



▲ Figure 13.2 Intermolecular interactions involved in solutions.

Three kinds of intermolecular interactions are involved in solution formation:

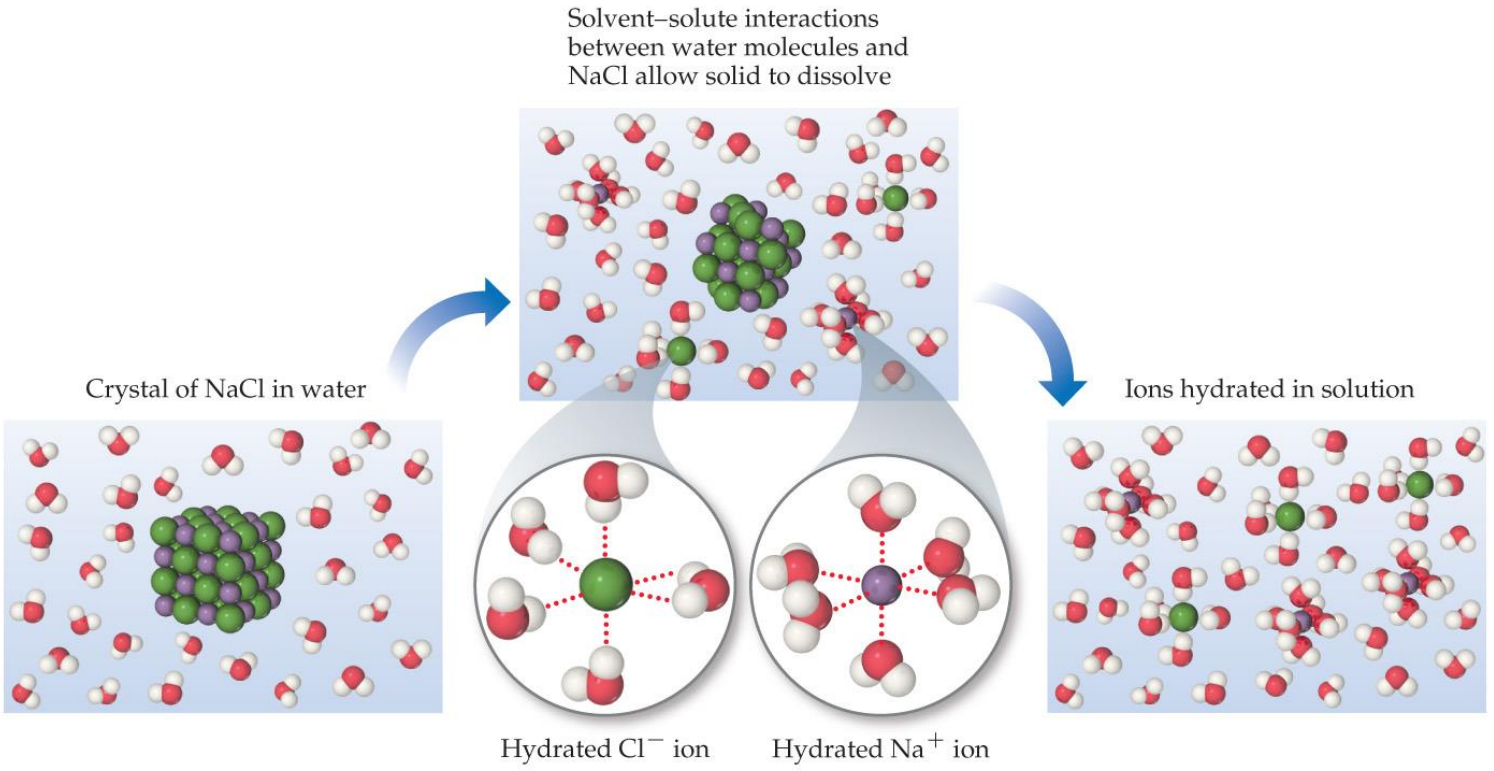
1. *Solute–solute* interactions between solute particles must be overcome to disperse the solute particles through the solvent.
2. *Solvent–solvent* interactions between solvent particles must be overcome to make room for the solute particles in the solvent.
3. *Solvent–solute* interactions between the solvent and solute particles occur as the particles mix.

The extent to which one substance is able to dissolve in another depends on the relative magnitudes of these three types of interactions. Solutions form when the magnitudes of the solvent–solute interactions are either comparable to or greater than the solute–solute and solvent–solvent interactions. For example, heptane (C_7H_{16}) and pentane (C_5H_{12}) dissolve in each other in all proportions. For this discussion, we can arbitrarily call heptane the solvent and pentane the solute. Both substances are nonpolar, and the magnitudes of the solvent–solute interactions (attractive dispersion forces) are comparable to the solute–solute and the solvent–solvent interactions. Thus, no forces impede mixing, and the tendency to mix (increase entropy) causes the solution to form spontaneously.

Solid NaCl dissolves readily in water because the attractive solvent–solute interactions between the polar H_2O molecules and the ions are strong enough to overcome the attractive solute–solute interactions between ions in the NaCl(s) and the attractive solvent– solvent interactions between H_2O molecules.

How Does a Solution Form?

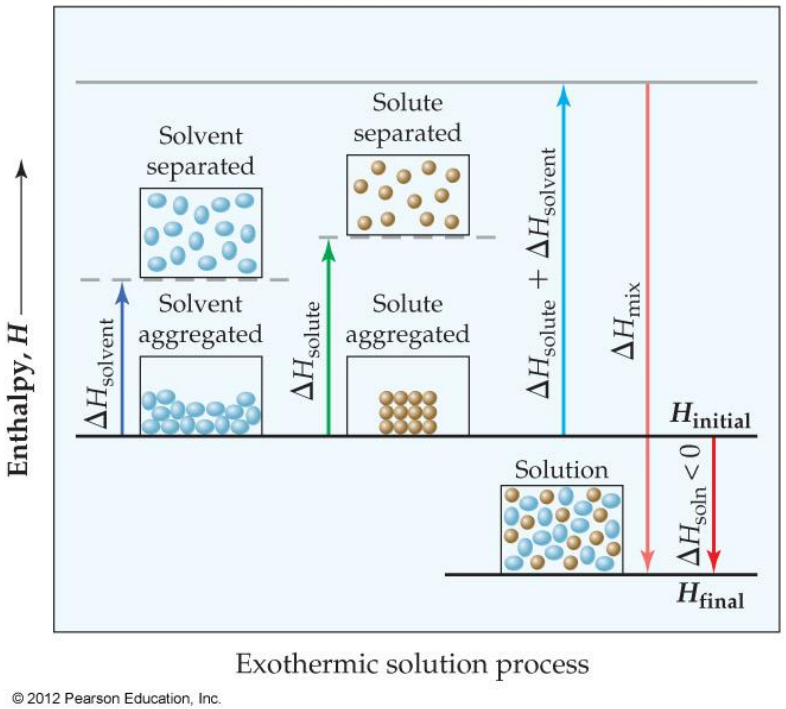
As a solution forms, the solvent pulls solute particles apart and surrounds, or **solvates**, them.



© 2012 Pearson Education, Inc.

Energy Changes in Solution

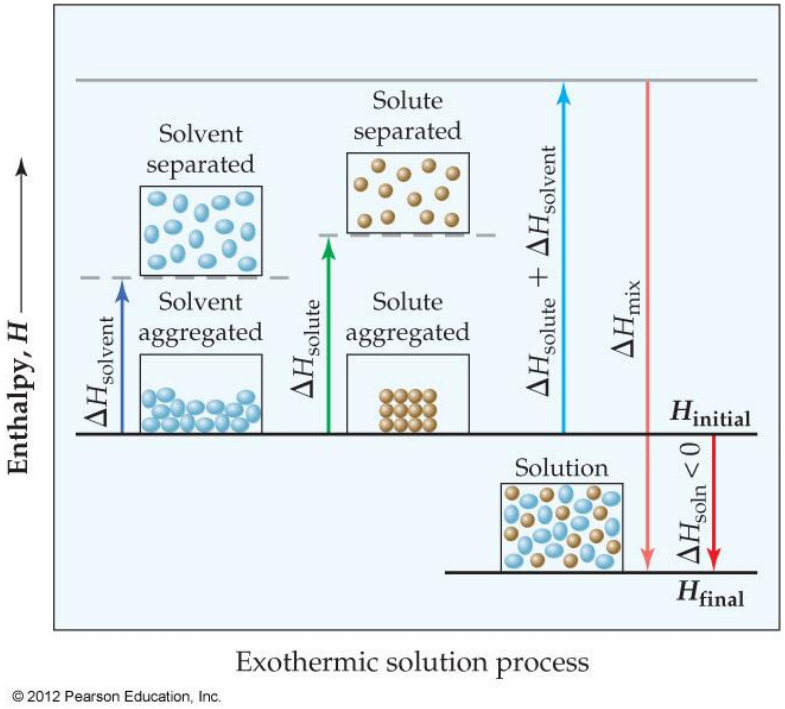
- Simply put, three processes affect the energetics of solution:
  - Separation of solute particles,
  - Separation of solvent particles,
  - New interactions between solute and solvent.



© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Energy Changes in Solution



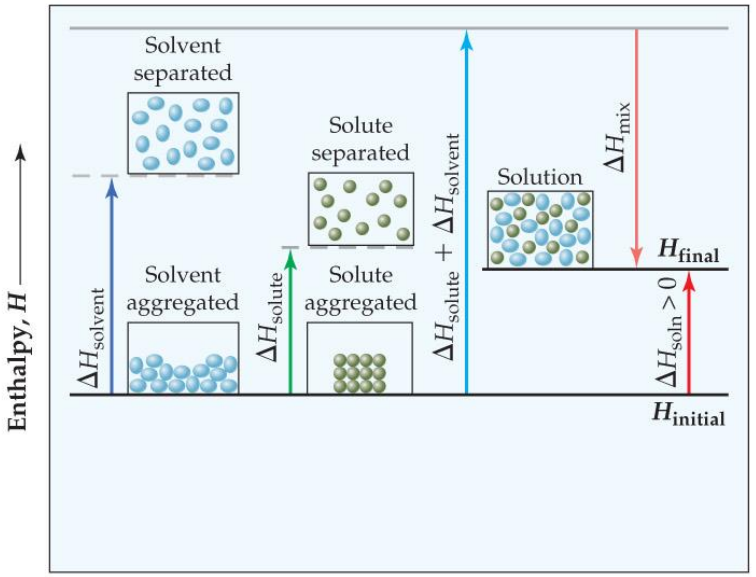
© 2012 Pearson Education, Inc.

The enthalpy change of the overall process depends on ΔH for each of these steps.

© 2012 Pearson Education, Inc.

Why Do Endothermic Processes Occur?

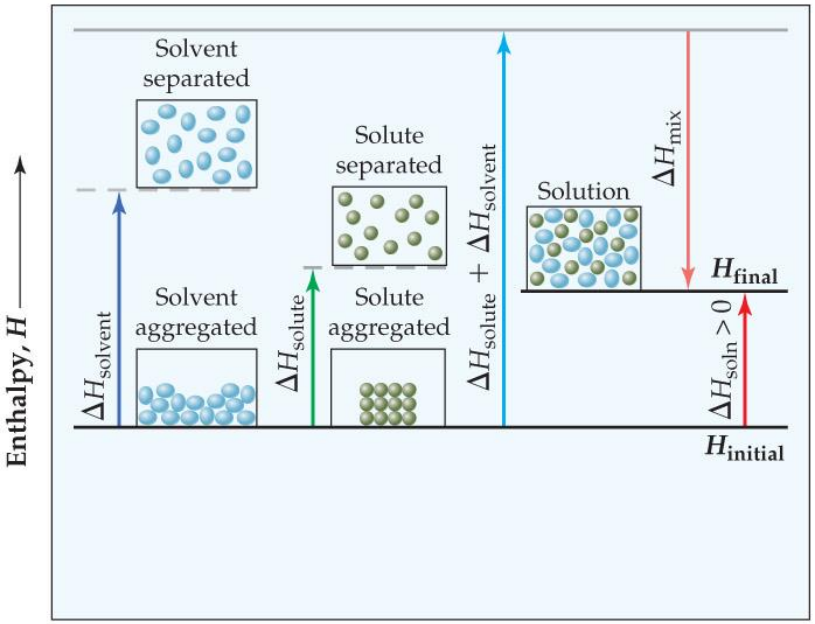
Things do not tend to occur spontaneously (i.e., without outside intervention) unless the energy of the system is lowered.



Endothermic solution process

© 2012 Pearson Education, Inc.

Why Do Endothermic Processes Occur?



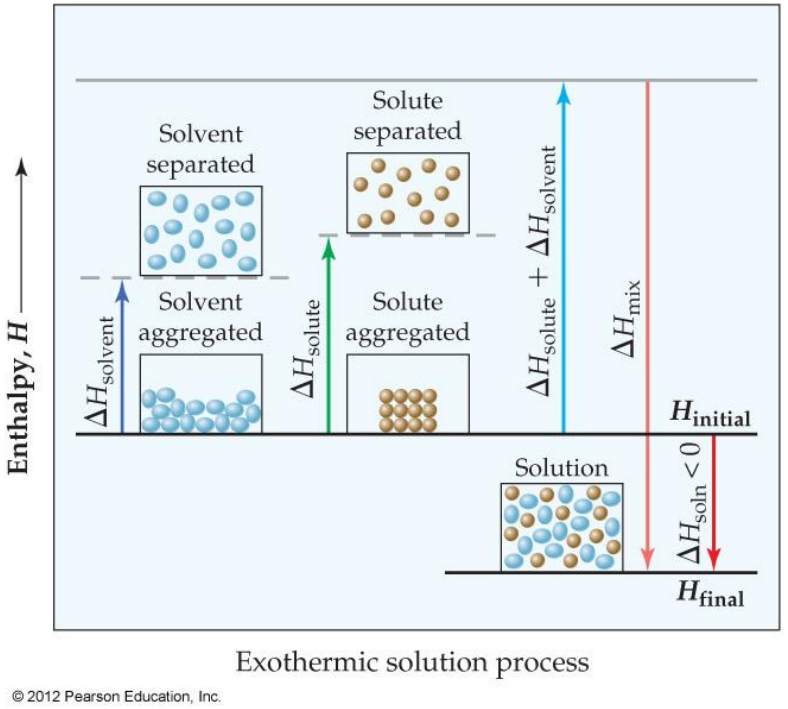
Endothermic solution process

Yet we know that in some processes, like the dissolution of NH_4NO_3 in water, heat is absorbed, not released.

© 2012 Pearson Education, Inc.

Enthalpy Is Only Part of the Picture

The reason is that increasing the disorder or randomness (known as **entropy**) of a system tends to lower the energy of the system.



© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

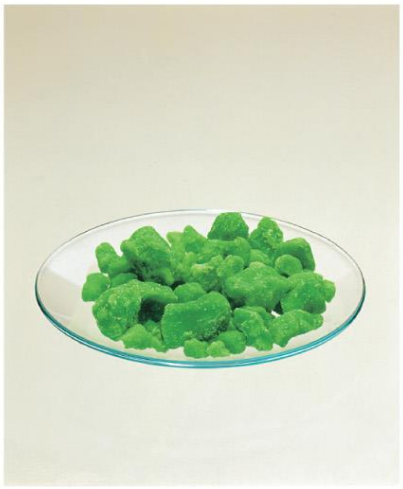
Solution Formation and Chemical Reactions



Nickel metal and hydrochloric acid



Nickel reacts with hydrochloric acid, forming $NiCl_2(aq)$ and $H_2(g)$. The solution is of $NiCl_2$, not Ni metal



$NiCl_2 \cdot 6H_2O(s)$ remains when solvent evaporated

© 2012 Pearson Education, Inc.

Just because a substance disappears when it comes in contact with a solvent, it doesn't mean the substance dissolved. It may have reacted.

© 2012 Pearson Education, Inc.

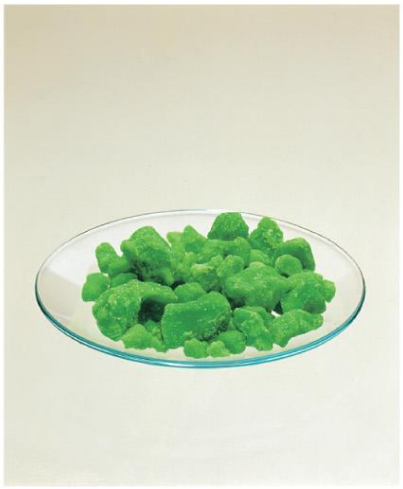
Student, Beware!



Nickel metal and hydrochloric acid



Nickel reacts with hydrochloric acid, forming $NiCl_2(aq)$ and $H_2(g)$. The solution is of $NiCl_2$, not Ni metal



$NiCl_2 \cdot 6H_2O(s)$ remains when solvent evaporated

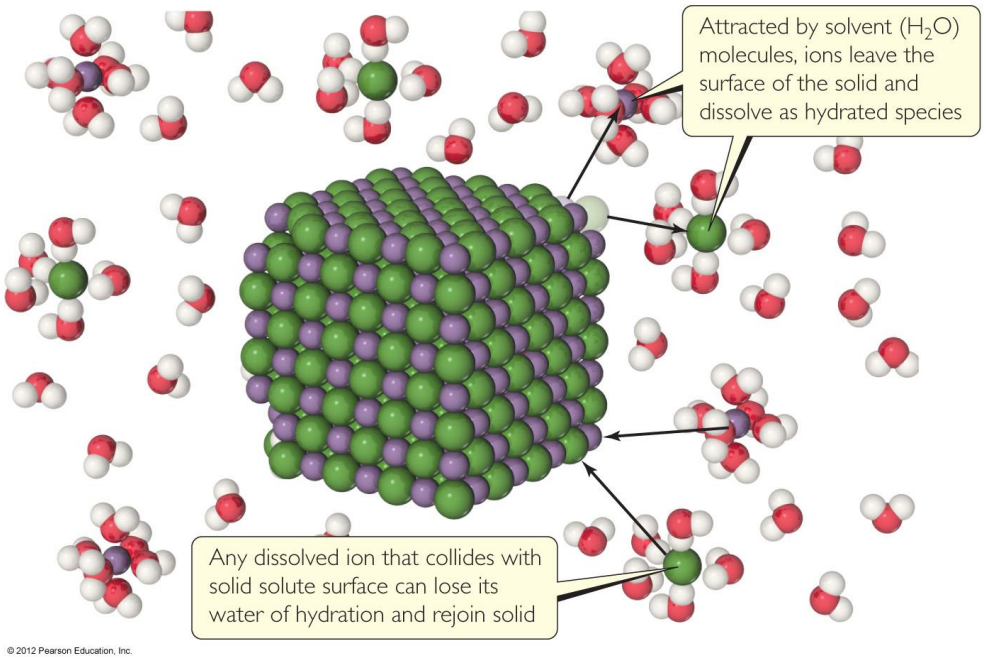
© 2012 Pearson Education, Inc.

- Dissolution is a physical change—you can get back the original solute by evaporating the solvent.
- If you can't get it back, the substance didn't dissolve, it reacted.

© 2012 Pearson Education, Inc.

When $NaCl_{(s)}$ is dissolved in water, on the other hand, no chemical reaction occurs. If the solution is evaporated to dryness, NaCl is recovered.

Types of Solutions

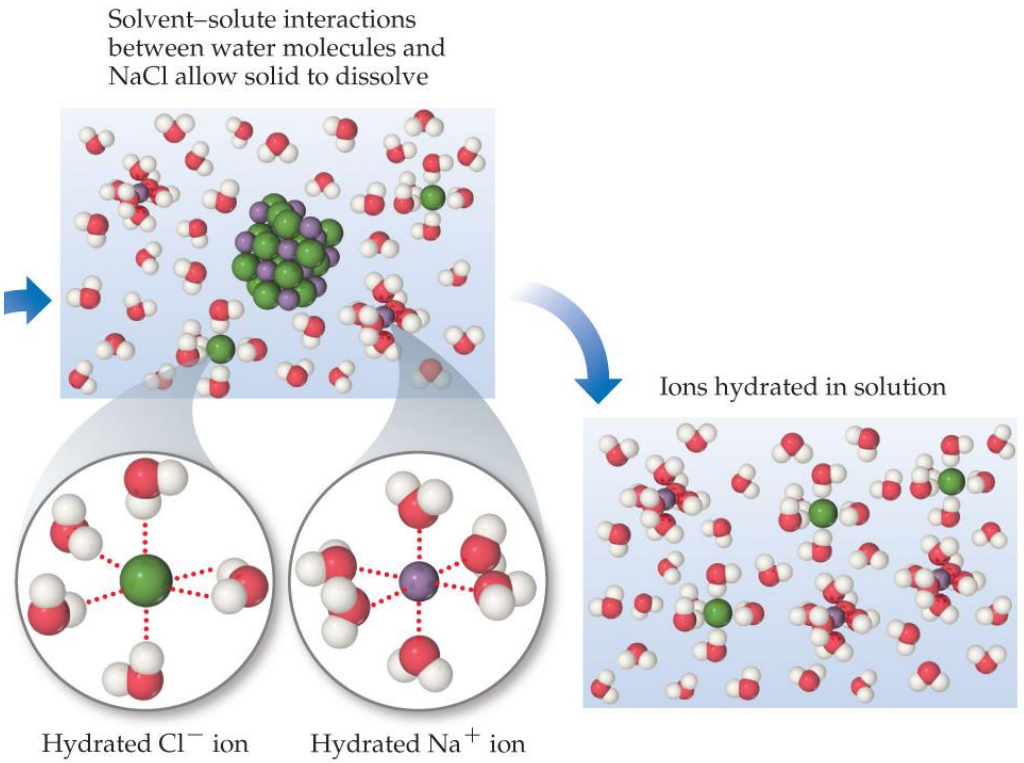


- Saturated
  - In a **saturated** solution, the solvent holds as much solute as is possible at that temperature.
  - Dissolved solute is in dynamic equilibrium with solid solute particles.

© 2012 Pearson Education, Inc.

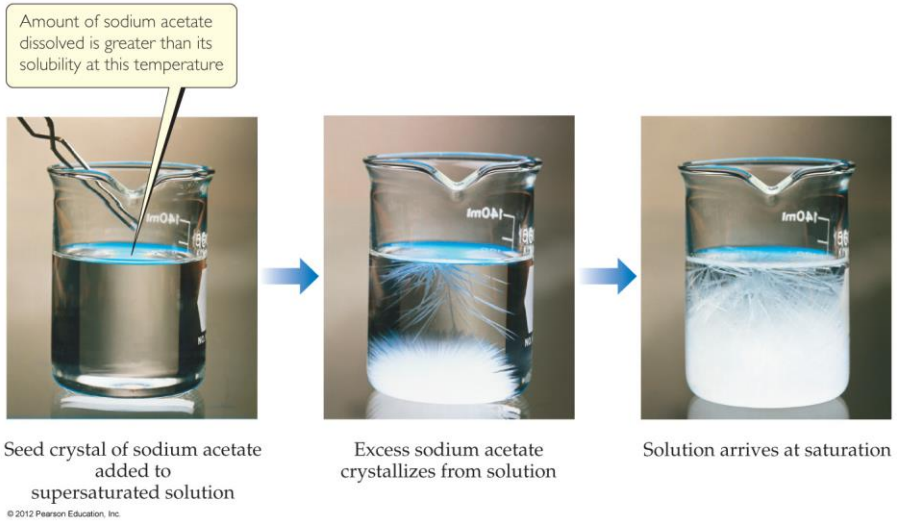
Types of Solutions

- Unsaturated
  - If a solution is **unsaturated**, less solute than can dissolve in the solvent at that temperature is dissolved in the solvent.



© 2012 Pearson Education, Inc.

Types of Solutions



- Supersaturated
  - In **supersaturated** solutions, the solvent holds more solute than is normally possible at that temperature.
  - These solutions are unstable; crystallization can usually be stimulated by adding a “seed crystal” or scratching the side of the flask.

© 2012 Pearson Education, Inc.

Factors Affecting Solubility

- Chemists use the axiom "like dissolves like."
  - Polar substances tend to dissolve in polar solvents.
  - Nonpolar substances tend to dissolve in nonpolar solvents.

TABLE 13.2 • Solubilities of Some Alcohols in Water and in Hexane*

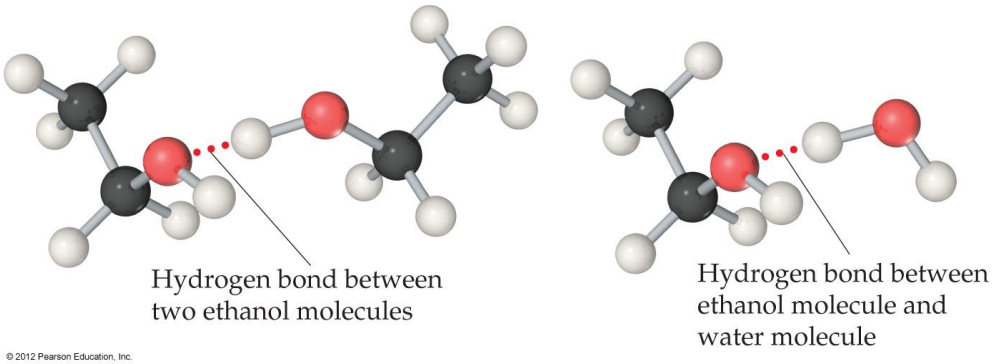
Alcohol	Solubility in H_2O	Solubility in C_6H_{14}
CH_3OH (methanol)	∞	0.12
CH_3CH_2OH (ethanol)	∞	∞
$CH_3CH_2CH_2OH$ (propanol)	∞	∞
$CH_3CH_2CH_2CH_2OH$ (butanol)	0.11	∞
$CH_3CH_2CH_2CH_2CH_2OH$ (pentanol)	0.030	∞
$CH_3CH_2CH_2CH_2CH_2CH_2OH$ (hexanol)	0.0058	∞

*Expressed in mol alcohol/100 g solvent at 20 °C. The infinity symbol (∞) indicates that the alcohol is completely miscible with the solvent.

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Factors Affecting Solubility



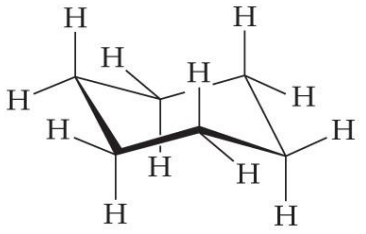
© 2012 Pearson Education, Inc.

The more similar the intermolecular attractions, the more likely one substance is to be soluble in another.

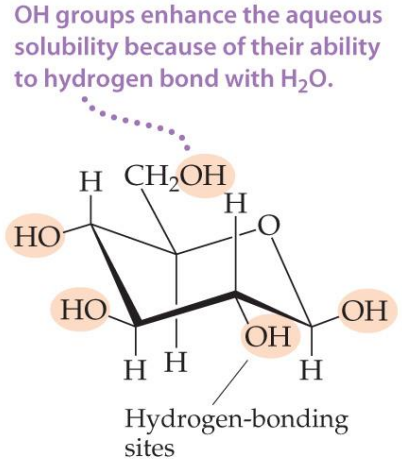
© 2012 Pearson Education, Inc.

Factors Affecting Solubility

Glucose (which has hydrogen bonding) is very soluble in water, while cyclohexane (which only has dispersion forces) is not.



Cyclohexane, C_6H_{12}, which has no polar OH groups, is essentially insoluble in water



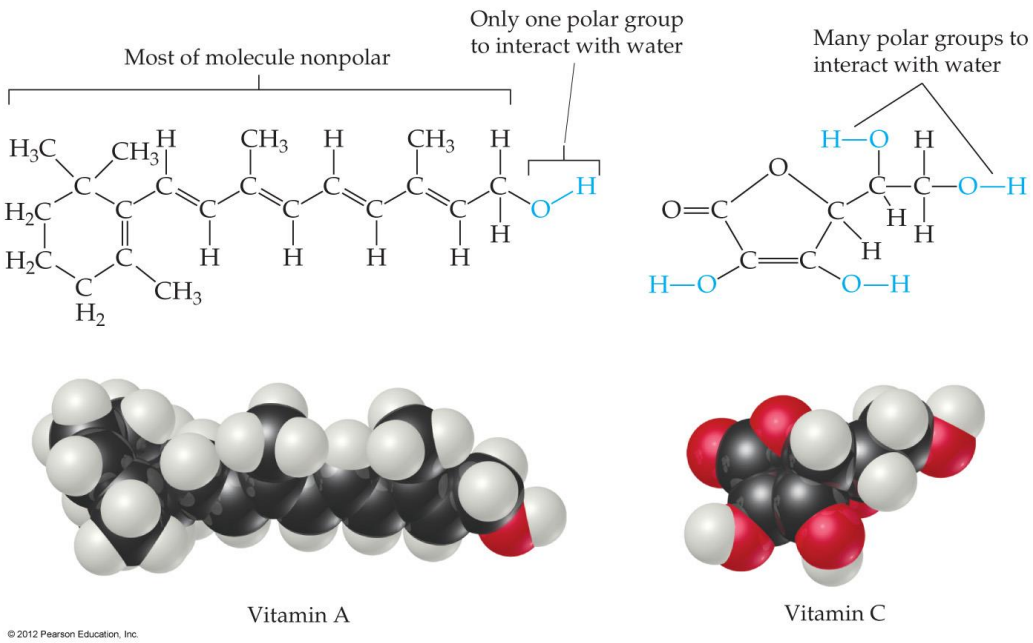
Glucose, $C_6H_{12}O_6$, has five OH groups and is highly soluble in water

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Factors Affecting Solubility

- Vitamin A is soluble in nonpolar compounds (like fats).
- Vitamin C is soluble in water.



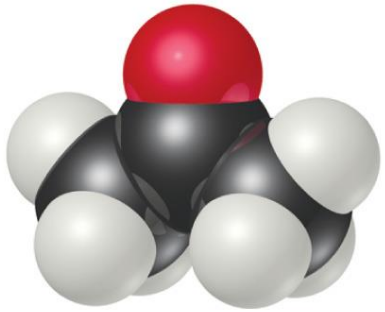
© 2012 Pearson Education, Inc.

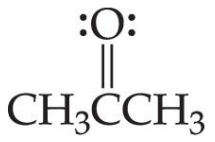
Gases in Solution

- In general, the solubility of gases in water increases with increasing mass.
- Larger molecules have stronger dispersion forces.

TABLE 13.1 • Solubilities of Gases in Water at 20 °C, with 1 atm Gas Pressure

Gas	Solubility (M)
N_2	0.69×10^{-3}
CO	1.04×10^{-3}
O_2	1.38×10^{-3}
Ar	1.50×10^{-3}
Kr	2.79×10^{-3}





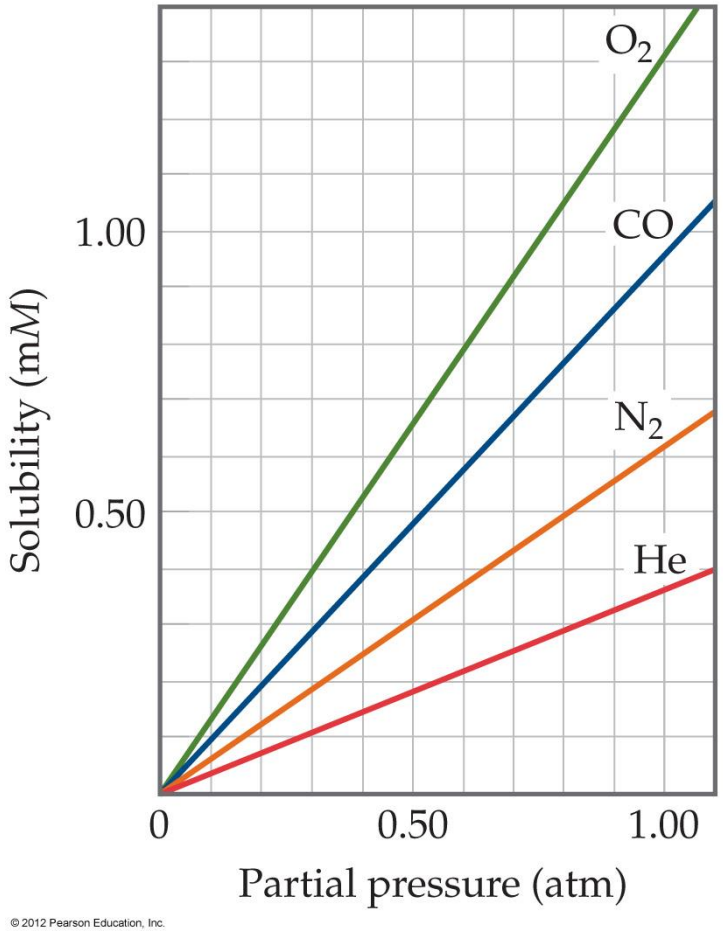
Acetone

© 2012 Pearson Education, Inc.

utions

© 2012 Pearson Education, Inc.

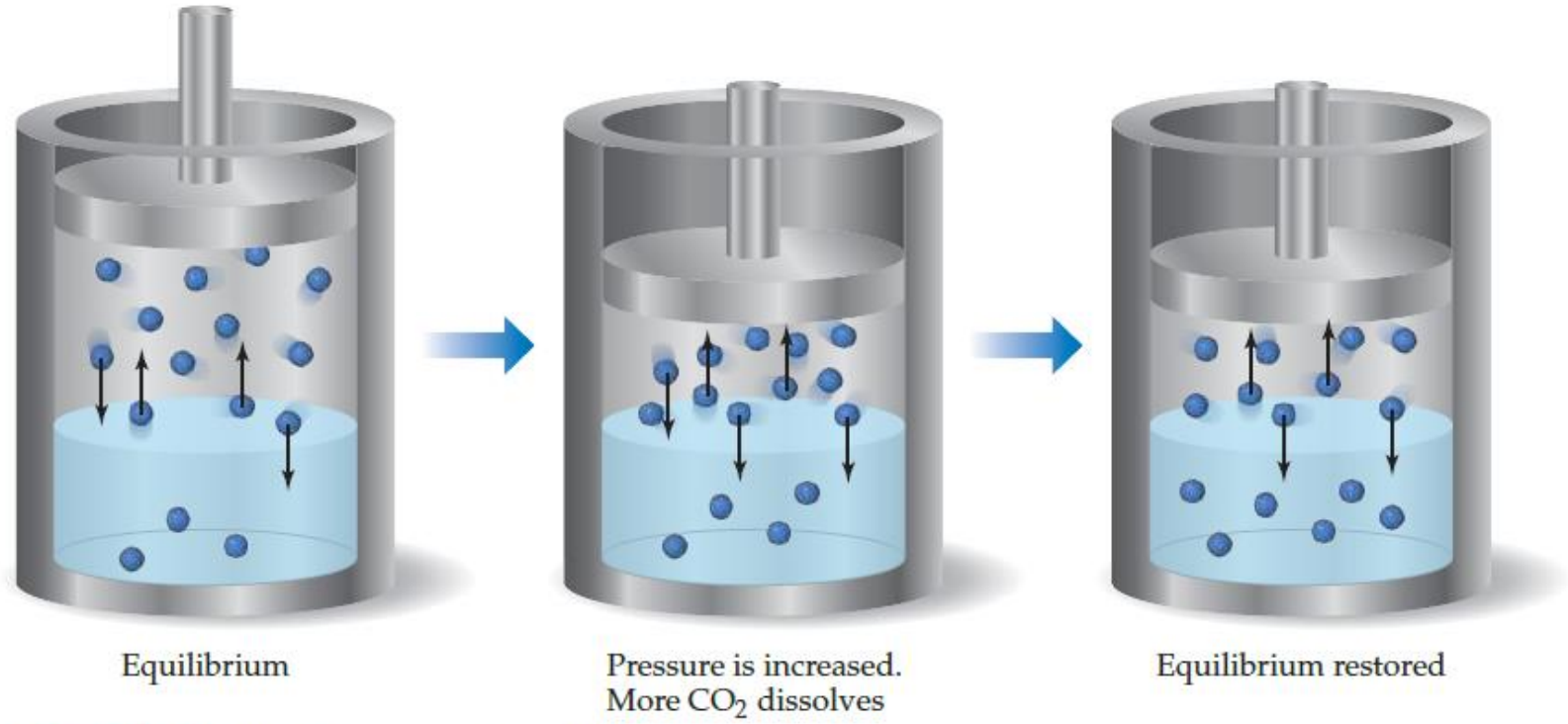
Gases in Solution



© 2012 Pearson Education, Inc.

- The solubility of liquids and solids does not change appreciably with pressure.
- But the solubility of a gas in a liquid is directly proportional to its pressure.

© 2012 Pearson Education, Inc.



▲ Figure 13.12 Effect of pressure on gas solubility.

Henry's Law

$$S_g = kP_g$$

where

- S_g is the solubility of the gas,
- k is the Henry's Law constant for that gas in that solvent, and
- P_g is the partial pressure of the gas above the liquid.

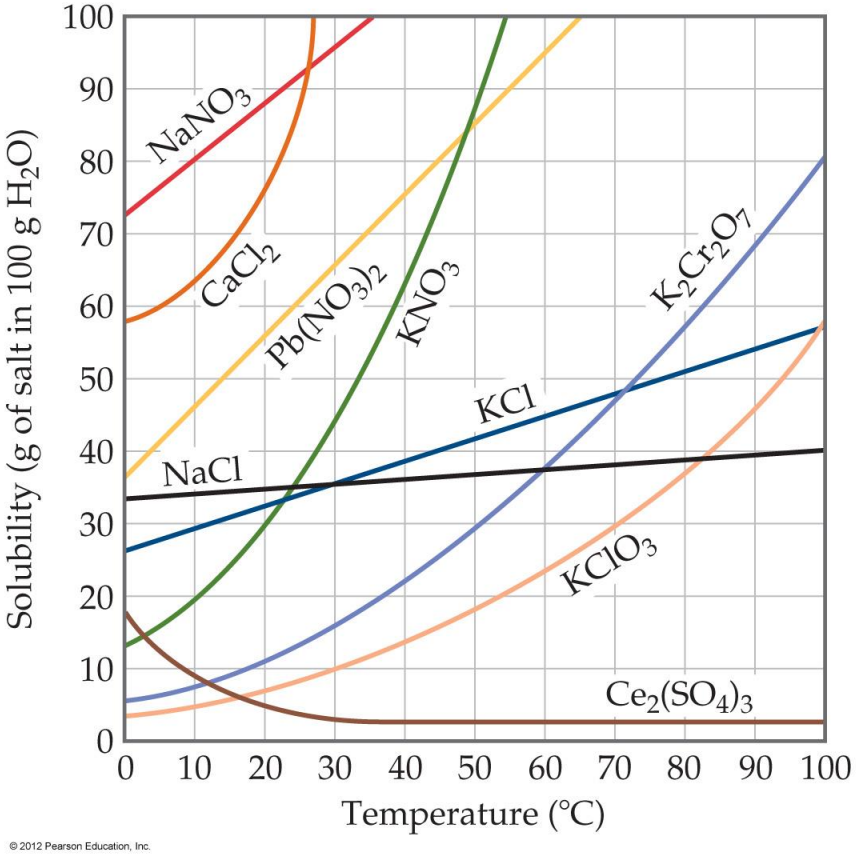


© 2012 Pearson Education, Inc.

Calculate the concentration of CO_2 in a soft drink that is bottled with a partial pressure of CO_2 of 4.0 atm over the liquid at 25 °C. The Henry's law constant for CO_2 in water at this temperature is 3.4×10^{-2} mol/L-atm.

$$S_{CO_2} = kP_{CO_2} = (3.4 \times 10^{-2}\,\text{mol/L-atm})(4.0\,\text{atm}) = 0.14\,\text{mol/L} = 0.14\,M$$

Temperature



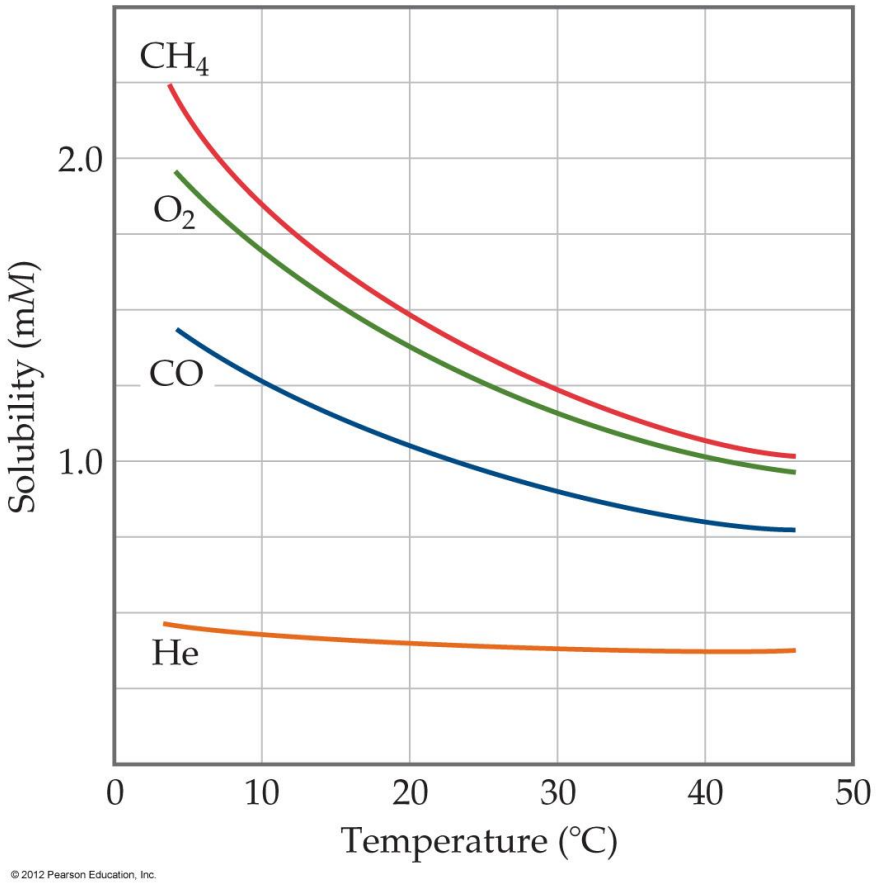
© 2012 Pearson Education, Inc.

Generally, the solubility of solid solutes in liquid solvents increases with increasing temperature.

© 2012 Pearson Education, Inc.

Temperature

- The opposite is true of gases.
  - Carbonated soft drinks are more "bubbly" if stored in the refrigerator.
  - Warm lakes have less O_2 dissolved in them than cool lakes.



© 2012 Pearson Education, Inc.

Ways of Expressing Concentrations of Solutions

© 2012 Pearson Education, Inc.

Mass Percentage

$$\text{Mass \% of } A = \frac{\text{mass of } A \text{ in solution}}{\text{total mass of solution}} \times 100$$

© 2012 Pearson Education, Inc.

Parts per Million and Parts per Billion

Parts per million (ppm)

$$\text{ppm} = \frac{\text{mass of } A \text{ in solution}}{\text{total mass of solution}} \times 10^6$$

Parts per billion (ppb)

$$\text{ppb} = \frac{\text{mass of } A \text{ in solution}}{\text{total mass of solution}} \times 10^9$$

© 2012 Pearson Education, Inc.

Mole Fraction (X)

$$X_A = \frac{\text{moles of } A}{\text{total moles of all components}}$$

- In some applications, one needs the mole fraction of *solvent*, not solute—make sure you find the quantity you need!

© 2012 Pearson Education, Inc.

Molarity (*M*)

$$M = \frac{\text{moles of solute}}{\text{liters of solution}}$$

- You will recall this concentration measure from Chapter 4.
- Since volume is temperature-dependent, molarity can change with temperature.

© 2012 Pearson Education, Inc.

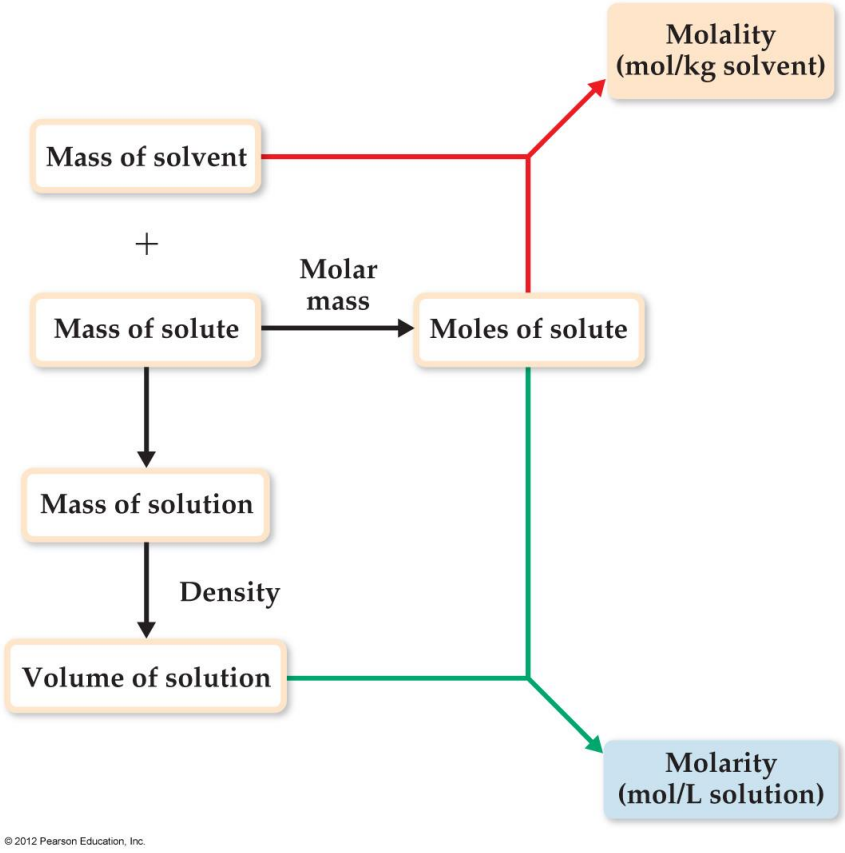
Molality (*m*)

$$m = \frac{\text{moles of solute}}{\text{kilograms of solvent}}$$

Since both moles and mass do not change with temperature, molality (unlike molarity) is *not* temperature-dependent.

© 2012 Pearson Education, Inc.

Changing Molarity to Molality



If we know the density of the solution, we can calculate the molality from the molarity, and vice versa.

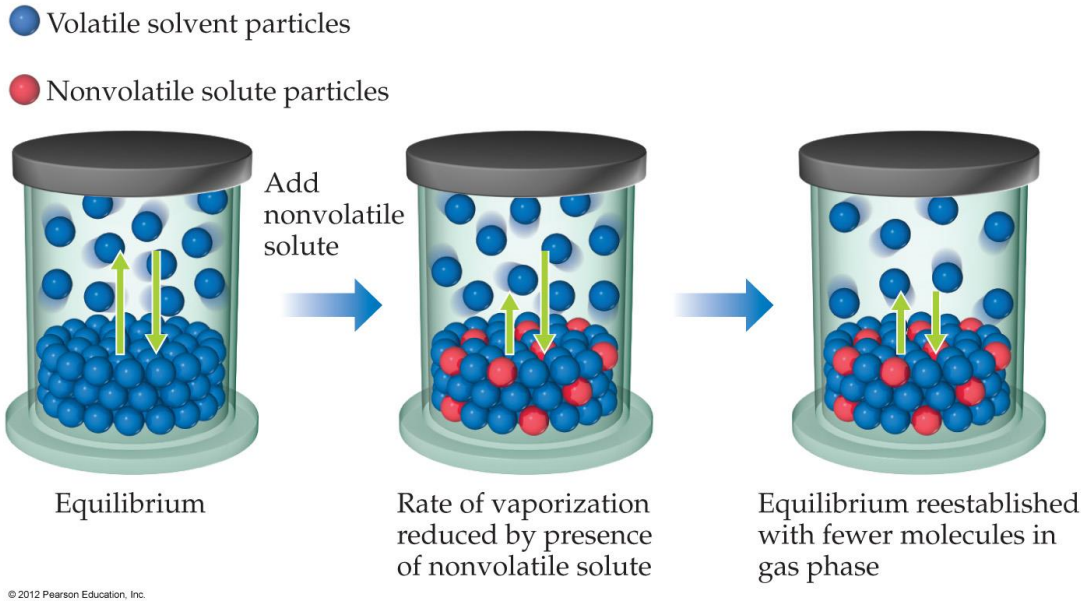
© 2012 Pearson Education, Inc.

Colligative Properties

- Changes in **colligative properties** depend only on the *number* of solute particles present, not on the *identity* of the solute particles.
- Among colligative properties are
  - Vapor-pressure lowering
  - Boiling-point elevation
  - Melting-point depression
  - Osmotic pressure

© 2012 Pearson Education, Inc.

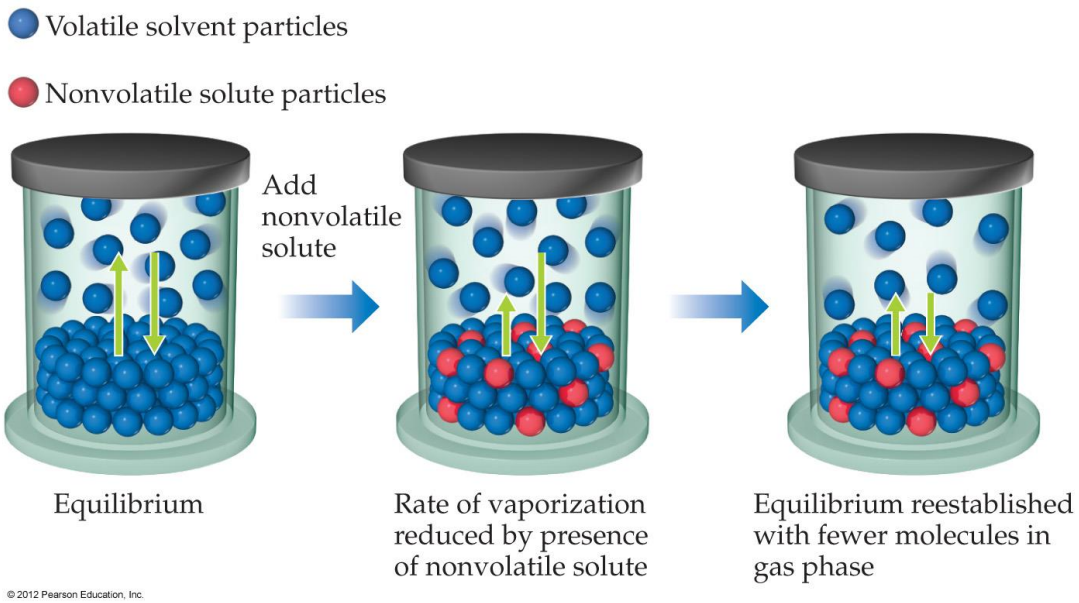
Vapor Pressure



Because of solute–solvent intermolecular attraction, higher concentrations of nonvolatile solutes make it harder for solvent to escape to the vapor phase.

© 2012 Pearson Education, Inc.

Vapor Pressure



Therefore, the vapor pressure of a solution is lower than that of the pure solvent.

© 2012 Pearson Education, Inc.

Raoult's Law

$$P_A = X_A P°_A$$

where

- X_A is the mole fraction of compound *A*, and
- $P°_A$ is the normal vapor pressure of *A* at that temperature.

Note: This is one of those times when you want to make sure you have the vapor pressure of the *solvent*.

© 2012 Pearson Education, Inc.

Ideally, the vapor pressure of a *volatile* solvent above a solution containing a *nonvolatile* solute is proportional to the solvent's concentration in the solution. This relationship is expressed quantitatively by **Raoult's law**, which states that the partial pressure exerted by solvent vapor above the solution, $P_{solution}$, equals the product of the mole fraction of the solvent, $X_{solvent}$, times the vapor pressure of the pure solvent, $P^{\circ}_{solvent}$:

$$P_{solution} = X_{solvent} P^{\circ}_{solvent} \qquad [13.10]$$

For example, the vapor pressure of pure water at 20 °C is $P^{\circ}_{H_2O} = 17.5$ torr. Imagine holding the temperature constant while adding glucose ($C_6H_{12}O_6$) to the water so that the mole fractions in the resulting solution are $X_{H_2O} = 0.800$ and $X_{C_6H_{12}O_6} = 0.200$. According to Equation 13.10, the vapor pressure of the water above this solution is 80.0% of that of pure water:

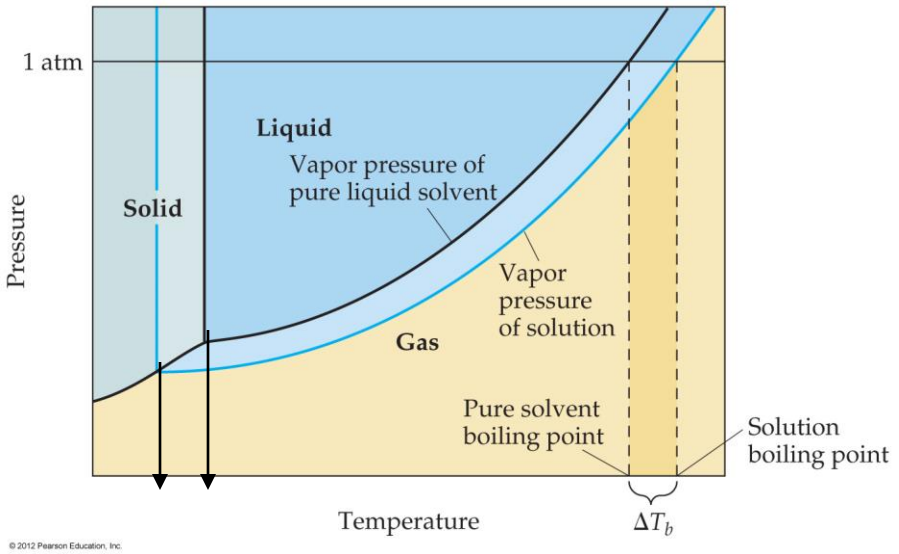
$$P_{solution} = (0.800)(17.5 \text{ torr}) = 14.0 \text{ torr}$$

The presence of the *nonvolatile* solute lowers the vapor pressure of the *volatile* solvent by 17.5 torr − 14.0 torr = 3.5 torr.

The vapor-pressure lowering caused by adding a nonvolatile solute depends on the total concentration of solute particles, regardless of whether they are molecules or ions.

Boiling-Point Elevation and Freezing-Point Depression

Nonvolatile solute–solvent interactions also cause solutions to have higher boiling points and lower freezing points than the pure solvent.



© 2012 Pearson Education, Inc.

Boiling-Point Elevation

TABLE 13.3 • Molal Boiling-Point-Elevation and Freezing-Point-Depression Constants

Solvent	Normal Boiling Point (°C)	K_b (°C/*m*)	Normal Freezing Point (°C)	K_f (°C/*m*)
Water, H_2O	100.0	0.51	0.0	1.86
Benzene, C_6H_6	80.1	2.53	5.5	5.12
Ethanol, C_2H_5OH	78.4	1.22	−114.6	1.99
Carbon tetrachloride, CCl_4	76.8	5.02	−22.3	29.8
Chloroform, $CHCl_3$	61.2	3.63	−63.5	4.68

© 2012 Pearson Education, Inc.

- The change in boiling point is proportional to the molality of the solution:

$$\Delta T_b = K_b \cdot m$$

where K_b is the molal boiling-point elevation constant, a property of the solvent.

ΔT_b is *added to* the normal boiling point of the solvent.

© 2012 Pearson Education, Inc.

Some typical values of K_b and K_f for several common solvents are given in ▲ Table 13.3. For water, the table shows $K_b = 0.51\ ^\circ C/m$, which means that the boiling point of any aqueous solution that is 1 m in nonvolatile solute particles is 0.51 °C higher than the boiling point of pure water. Because solutions generally do

The increase in the boiling point of a solution, relative to the pure solvent, depends on the molality of the solute. But it is important to remember that boiling-point elevation is proportional to the *total* concentration of solute particles, regardless of whether the particles are molecules or ions. When NaCl dissolves in water, 2 mol of solute particles (1 mol of Na^+ and 1 mol of Cl^-) are formed for each mole of NaCl that dissolves. We take this into account by defining i, the **van't Hoff factor**, as the number of fragments that a solute breaks up into for a particular solvent. The change in boiling point for a solution compared to the pure solvent is:

$$\Delta T_b = T_b(\text{solution}) - T_b(\text{solvent}) = iK_bm \qquad [13.12]$$

In this equation, T_b(solution) is the boiling point of the solution, T_b(solvent) is the boiling point of the pure solvent, m is the molality of the solute, K_b is the **molal boiling-point-elevation constant** for the solvent (which is a proportionality constant that is experimentally determined for each solvent), and i is the van't Hoff factor. For a nonelectrolyte, we can always assume $i = 1$; for an electrolyte, i will depend on how the substance ionizes in that solvent. For instance, $i = 2$ for NaCl in water, assuming complete dissociation of ions. As a result, we expect the boiling-point elevation of a 1 m aqueous solution of NaCl to be twice as large as the boiling-point elevation of a 1 m solution of a nonelectrolyte such as sucrose. Thus, to properly predict the effect of a particular solute on boiling-point elevation (or any other colligative property), it is important to know whether the solute is an electrolyte or a nonelectrolyte. (Sections 4.1 and 4.3)

Freezing-Point Depression

TABLE 13.3 • Molal Boiling-Point-Elevation and Freezing-Point-Depression Constants

Solvent	Normal Boiling Point (°C)	K_b (°C/*m*)	Normal Freezing Point (°C)	K_f (°C/*m*)
Water, H_2O	100.0	0.51	0.0	1.86
Benzene, C_6H_6	80.1	2.53	5.5	5.12
Ethanol, C_2H_5OH	78.4	1.22	−114.6	1.99
Carbon tetrachloride, CCl_4	76.8	5.02	−22.3	29.8
Chloroform, $CHCl_3$	61.2	3.63	−63.5	4.68

© 2012 Pearson Education, Inc.

- The change in freezing point can be found similarly:

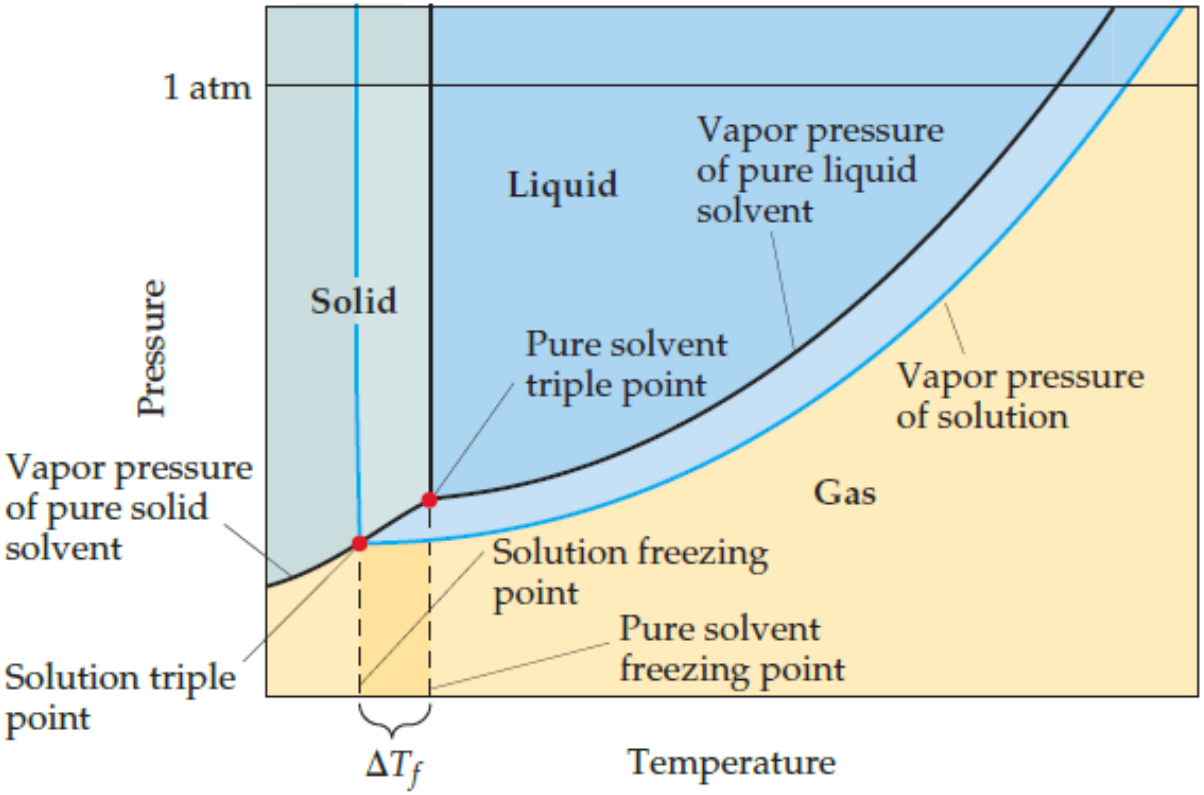
$$\Delta T_f = -K_f \bullet m$$

- Here K_f is the molal freezing-point depression constant of the solvent.

ΔT_f is *subtracted from* the normal freezing point of the solvent.

© 2012 Pearson Education, Inc.

Freezing-Point Depression



Like the boiling-point elevation, the change in freezing point ΔT_f is directly proportional to solute molality, taking into account the van't Hoff factor i:

$$\Delta T_f = T_f(\text{solution}) - T_f(\text{solvent}) = -iK_f m \qquad [13.13]$$

The proportionality constant K_f is the **molal freezing-point-depression constant**, analogous to K_b for boiling point elevation. Note that because the solution freezes at a *lower* temperature than does the pure solvent, the value of ΔT_f is *negative.*

For water, K_f is 1.86 °C/*m*. Therefore, any aqueous solution that is 1 *m* in nonvolatile solute particles (such as 1 *m* $C_6H_{12}O_6$ or 0.5 *m* NaCl) freezes at the temperature that is 1.86 °C lower than the freezing point of pure water.

The freezing-point depression caused by solutes has useful applications: it is why antifreeze works in car cooling systems, and why calcium chloride ($CaCl_2$) promotes the melting of ice on roads during winter.

Boiling-Point Elevation and Freezing-Point Depression

Note that in both equations, ΔT does not depend on *what the solute is*, but only on *how many particles* are dissolved.

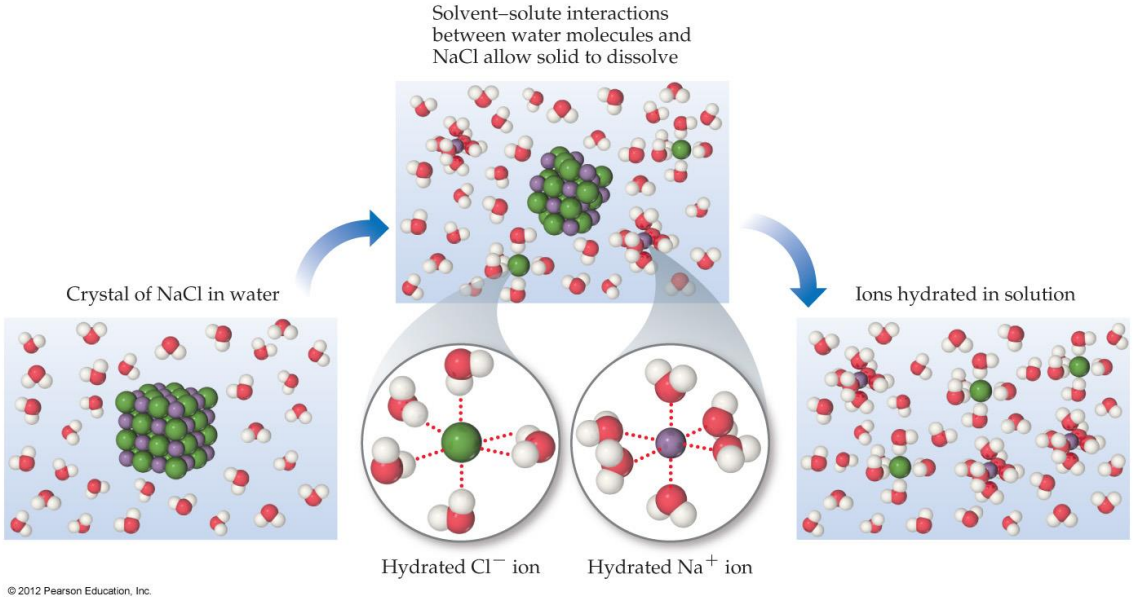
$$\Delta T_b = K_b \bullet m$$

$$\Delta T_f = -K_f \bullet m$$

© 2012 Pearson Education, Inc.

Colligative Properties of Electrolytes

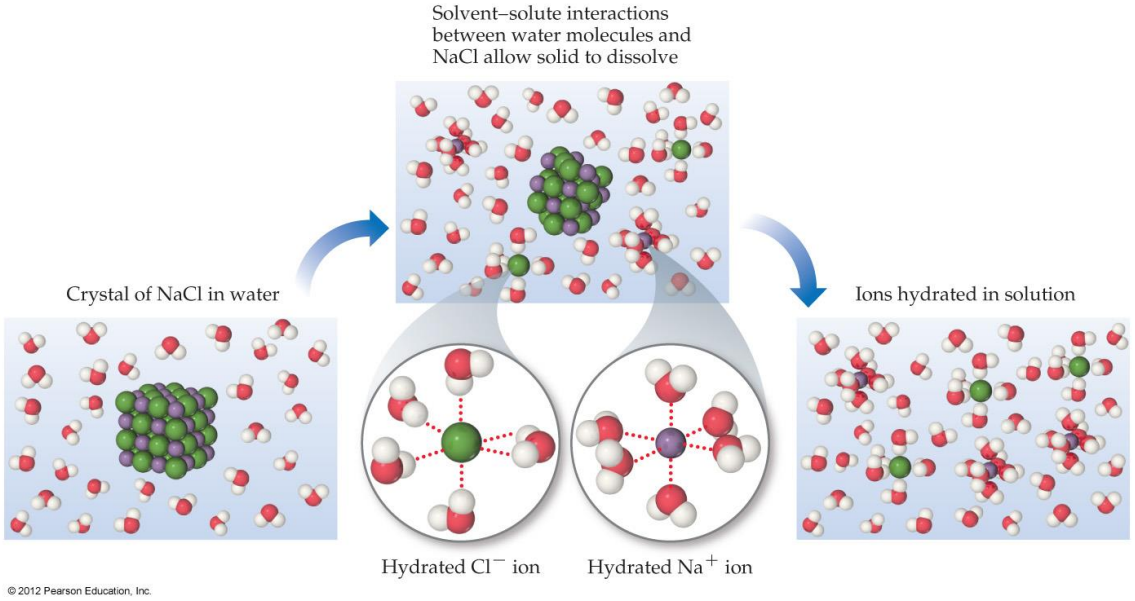
Since the colligative properties of electrolytes depend on the number of particles dissolved, solutions of electrolytes (which dissociate in solution) should show greater changes than those of nonelectrolytes.



© 2012 Pearson Education, Inc.

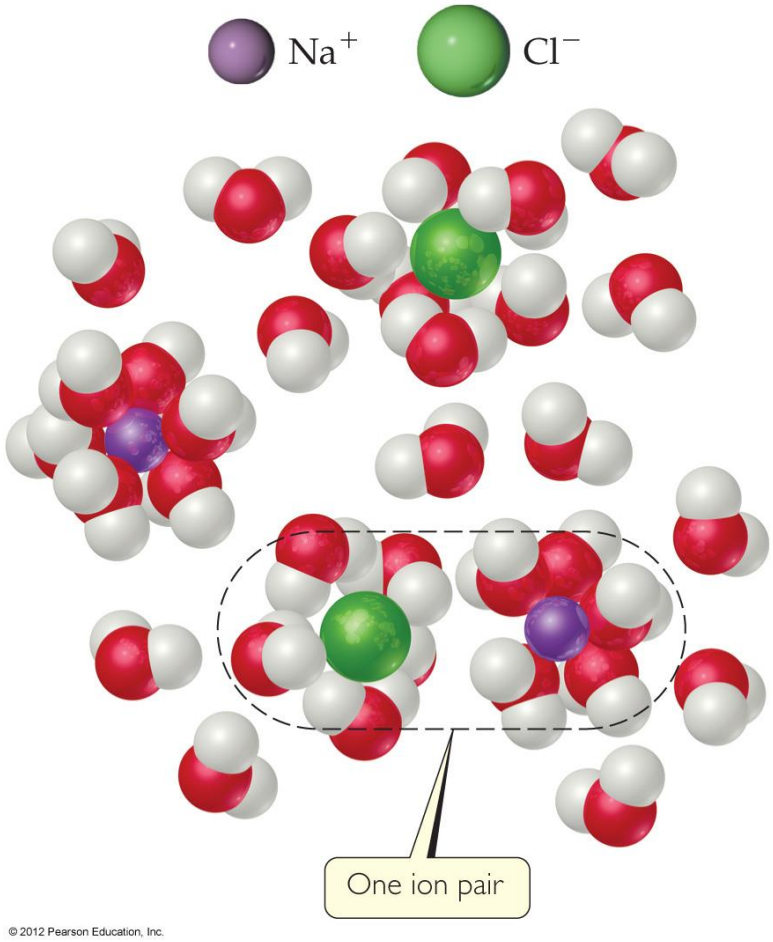
Colligative Properties of Electrolytes

However, a 1*M* solution of NaCl does not show twice the change in freezing point that a 1*M* solution of methanol does.



© 2012 Pearson Education, Inc.

van’t Hoff Factor

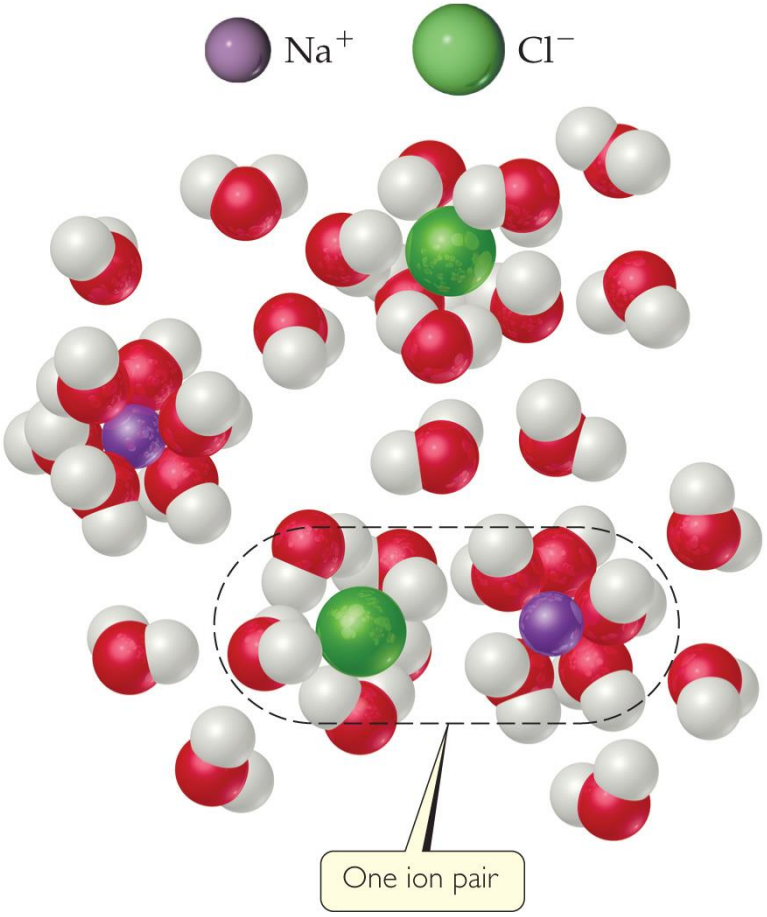


© 2012 Pearson Education, Inc.

One mole of NaCl in water does not really give rise to two moles of ions.

© 2012 Pearson Education, Inc.

van't Hoff Factor



© 2012 Pearson Education, Inc.

Some Na^+ and Cl^- reassociate for a short time, so the true concentration of particles is somewhat less than two times the concentration of NaCl.

© 2012 Pearson Education, Inc.

van’t Hoff Factor

- Reassociation is more likely at higher concentration.
- Therefore, the number of particles present is concentration-dependent.

TABLE 13.4 • van’t Hoff Factors for Several Substances at 25 °C

Compound	Concentration			Limiting Value
	0.100 *m*	0.0100 *m*	0.00100 *m*	
Sucrose	1.00	1.00	1.00	1.00
NaCl	1.87	1.94	1.97	2.00
K_2SO_4	2.32	2.70	2.84	3.00
$MgSO_4$	1.21	1.53	1.82	2.00

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

van't Hoff Factor

- We modify the previous equations by multiplying by the van't Hoff factor, *i:*

$$\Delta T_f = -K_f \bullet m \bullet i$$

TABLE 13.4 • van't Hoff Factors for Several Substances at 25 °C

Compound	Concentration			Limiting Value
	0.100 *m*	0.0100 *m*	0.00100 *m*	
Sucrose	1.00	1.00	1.00	1.00
NaCl	1.87	1.94	1.97	2.00
K_2SO_4	2.32	2.70	2.84	3.00
$MgSO_4$	1.21	1.53	1.82	2.00

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Automotive antifreeze contains ethylene glycol, $CH_2(OH)CH_2(OH)$, a nonvolatile nonelectrolyte, in water. Calculate the boiling point and freezing point of a 25.0% by mass solution of ethylene glycol in water.

$$\text{Molality} = \frac{\text{moles } C_2H_6O_2}{\text{kilograms } H_2O} = \left(\frac{250 \text{ g } C_2H_6O_2}{750 \text{ g } H_2O}\right)\left(\frac{1 \text{ mol } C_2H_6O_2}{62.1 \text{ g } C_2H_6O_2}\right)\left(\frac{1000 \text{ g } H_2O}{1 \text{ kg } H_2O}\right)$$
$$= 5.37\ m$$

$$\Delta T_b = iK_b m = (1)(0.51\,^\circ\text{C}/m)(5.37\ m) = 2.7\,^\circ\text{C}$$
$$\Delta T_f = -iK_f m = -(1)(1.86\,^\circ\text{C}/m)(5.37\ m) = -10.0\,^\circ\text{C}$$

Hence, the boiling and freezing points of the solution are readily calculated:

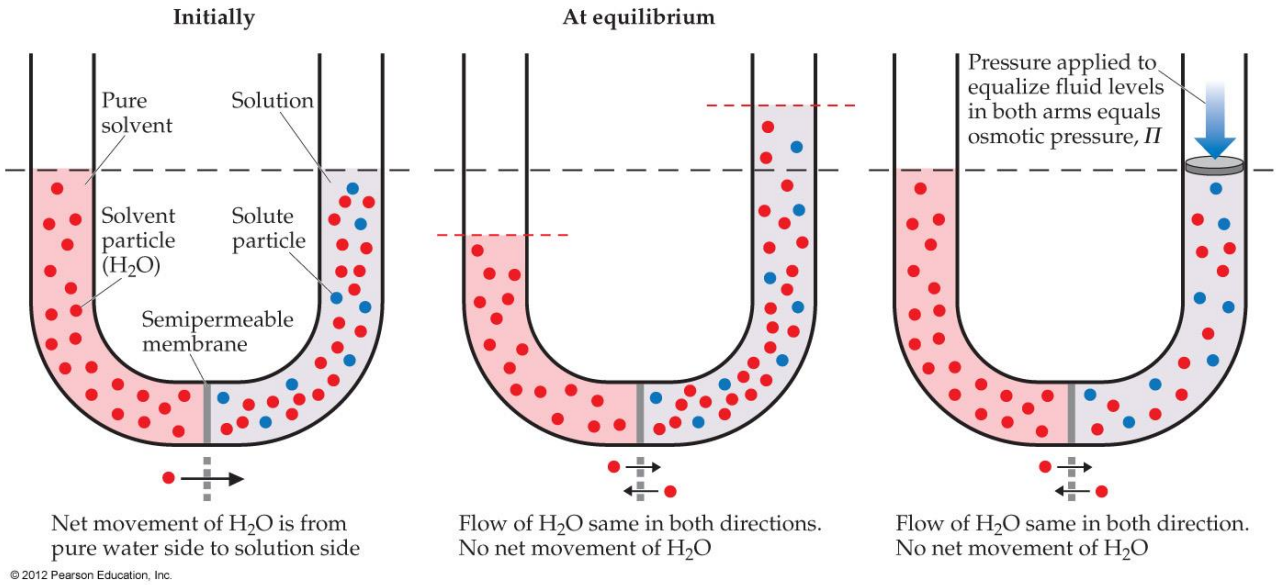
$$\Delta T_b = T_b(\text{solution}) - T_b(\text{solvent})$$
$$2.7\,^\circ\text{C} = T_b(\text{solution}) - 100.0\,^\circ\text{C}$$
$$T_b(\text{solution}) = 102.7\,^\circ\text{C}$$
$$\Delta T_f = T_f(\text{solution}) - T_f(\text{solvent})$$
$$-10.0\,^\circ\text{C} = T_f(\text{solution}) - 0.0\,^\circ\text{C}$$
$$T_f(\text{solution}) = -10.0\,^\circ\text{C}$$

Osmosis

- Some substances form **semipermeable membranes**, allowing some smaller particles to pass through, but blocking other larger particles.
- In biological systems, most semipermeable membranes allow water to pass through, but solutes are not free to do so.

© 2012 Pearson Education, Inc.

Osmosis



In osmosis, there is net movement of solvent from the area of **higher solvent concentration** (*lower solute concentration*) to the area of **lower solvent concentration** (*higher solute concentration*).

© 2012 Pearson Education, Inc.

Osmotic Pressure

The pressure required to stop osmosis, known as **osmotic pressure**, π, is

$$\pi = \left(\frac{n}{V}\right)RT = MRT$$

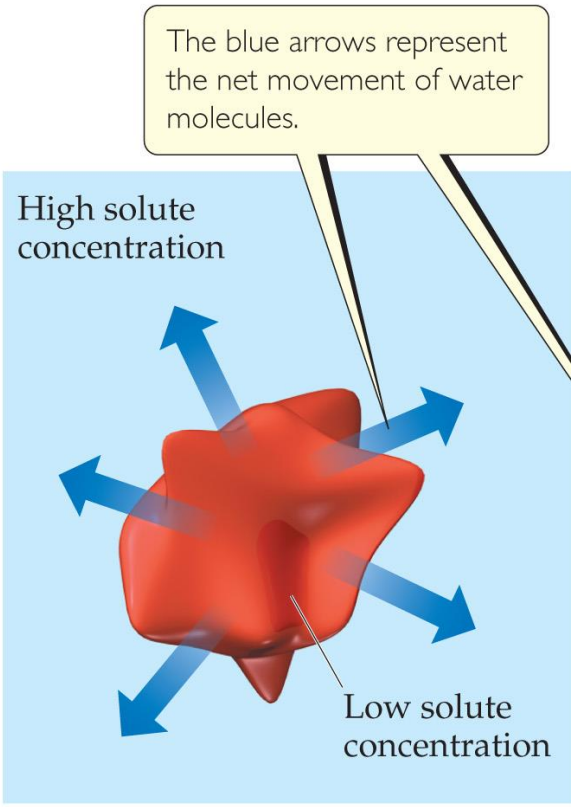
where *M* is the molarity of the solution.

If the osmotic pressure is the same on both sides of a membrane (i.e., the concentrations are the same), the solutions are **isotonic**.

© 2012 Pearson Education, Inc.

Osmosis in Blood Cells

- If the solute concentration outside the cell is greater than that inside the cell, the solution is **hypertonic**.
- Water will flow out of the cell, and **crenation** results.



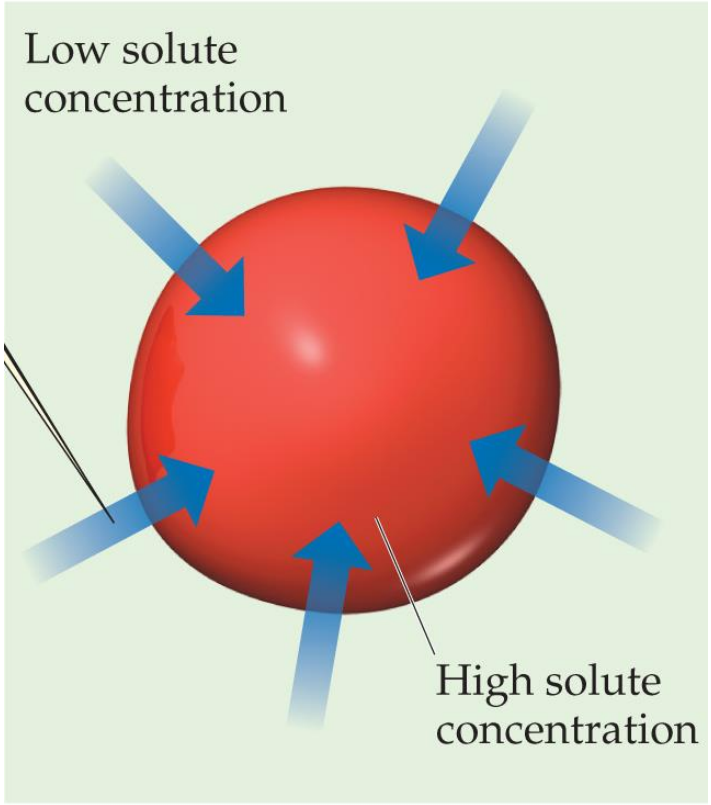
Crenation of red blood cell placed in hypertonic environment

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Osmosis in Cells

- If the solute concentration outside the cell is less than that inside the cell, the solution is **hypotonic**.
- Water will flow into the cell, and **hemolysis** results.



Hemolysis of red blood cell placed in hypotonic environment

© 2012 Pearson Education, Inc.

Colloids

Suspensions of particles larger than individual ions or molecules, but too small to be settled out by gravity, are called **colloids**.

TABLE 13.5 • Types of Colloids

Phase of Colloid	Dispersing (solvent-like) Substance	Dispersed (solute-like) Substance	Colloid Type	Example
Gas	Gas	Gas	—	None (all are solutions)
Gas	Gas	Liquid	Aerosol	Fog
Gas	Gas	Solid	Aerosol	Smoke
Liquid	Liquid	Gas	Foam	Whipped cream
Liquid	Liquid	Liquid	Emulsion	Milk
Liquid	Liquid	Solid	Sol	Paint
Solid	Solid	Gas	Solid foam	Marshmallow
Solid	Solid	Liquid	Solid emulsion	Butter
Solid	Solid	Solid	Solid sol	Ruby glass

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Tyndall Effect



- Colloidal suspensions can scatter rays of light.
- This phenomenon is known as the **Tyndall effect**.



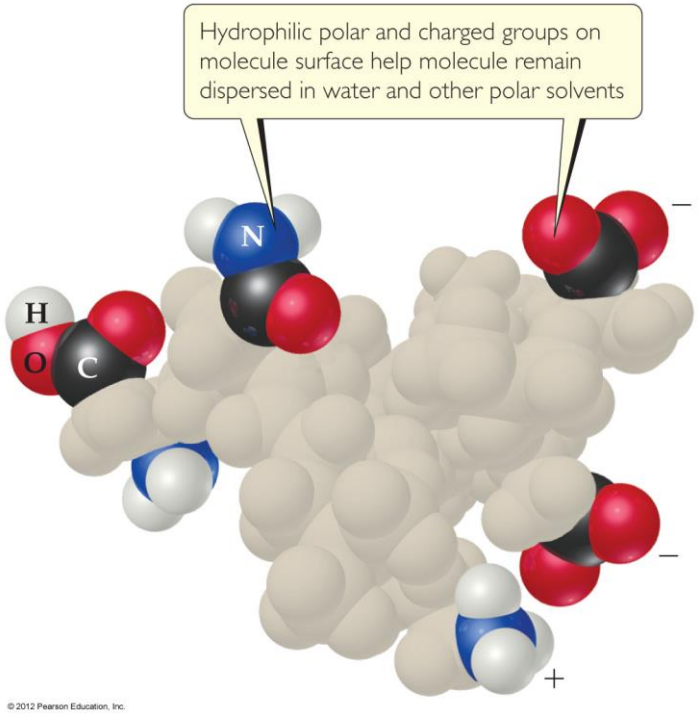


© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Colloids in Biological Systems

Some molecules have a polar, **hydrophilic** (*water-loving*) end and a nonpolar, **hydrophobic** (*water-hating*) end.

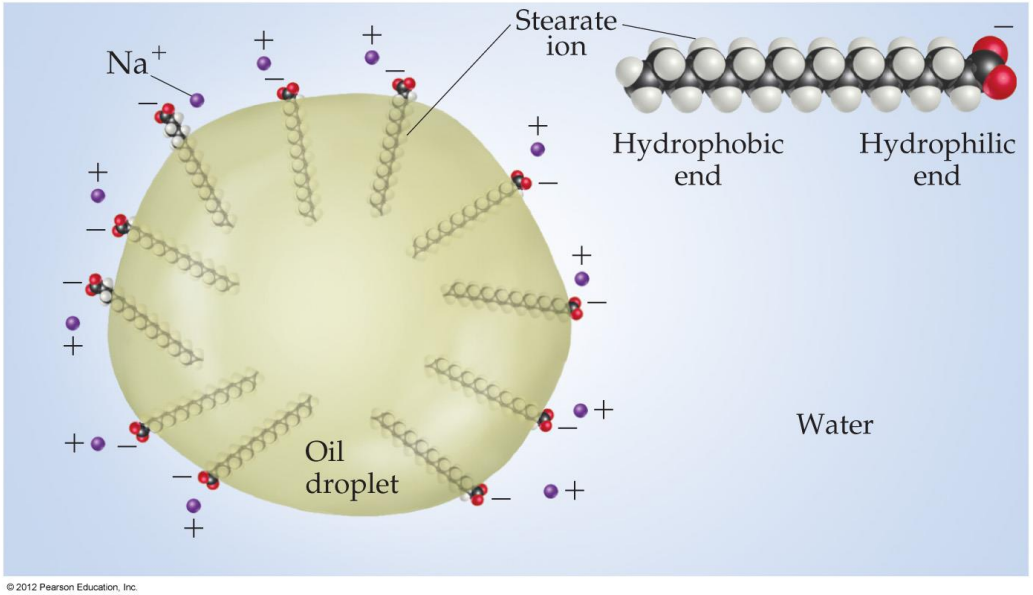


Solutions

© 2012 Pearson Education, Inc.

Colloids in Biological Systems

These molecules can aid in the emulsification of fats and oils in aqueous solutions.



© 2012 Pearson Education, Inc.